

Reimplementation of Show, Attend, Tell

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1 Introduction

Image captioning is challenging because it requires the model to identify salient objects in an image and reason about their relationships, then compress large amounts of visual information into concise human language. Previous work such as Show and Tell [1] approached this task using an encoder-decoder framework, where a CNN encodes the image into a fixed vector representation that is then passed to a RNN to generate the caption. However, representing the image as a single static feature vector from the convnet creates an information bottleneck.

Show, Attend and Tell [2] addresses this limitation by introducing an attention mechanism into the encoder-decoder framework, enabling the decoder to dynamically focus on relevant spatial regions while producing each word. Xu et al. present two variants: **soft attention**, which is fully differentiable and trained via backpropagation; and **hard attention**, which samples discrete spatial locations and is trained using REINFORCE. The paper demonstrates improved performance over non-attention baselines on Flickr8k, Flickr30k, and MS-COCO.

2 Chosen Result

Due to compute constraints, we only evaluate on the Flickr8k dataset [3]. We reproduce the quantitative results reported in the original paper (Table 2), focusing on BLEU-4 and METEOR (we include BLEU-1 through BLEU-3 for completeness in our poster). BLEU measures the exact n-gram overlap between generated and reference captions, while METEOR accounts for synonymy and stemming [4]. These metrics quantify caption quality and serve as primary evidence in the original paper that attention mechanisms outperform prior captioning systems. We also reproduce the qualitative attention visualizations, which demonstrate the interpretability advantage of the attention mechanism: where the model is “looking” at each timestep can be overlaid on the input image to explain which region drove the model to generate each predicted word. Beyond the paper, we replace the original VGG-19 encoder with ResNet-50 [5] to evaluate the impact of a stronger feature extractor.

3 Methodology

The attention-based encoder-decoder framework consists of a CNN encoder that extracts features and an LSTM decoder that generates captions using an attention mechanism over these features.

3.1 Encoder

We encode the input image using a pretrained CNN. Formally, the encoder outputs $a = \{a_1, \dots, a_L\}$, where each

$a_i \in \mathbb{R}^D$ represents a feature vector at a particular spatial location and L is the number of locations. Following the paper, we use VGG-19 to extract a 14×14 feature map where $a_i \in \mathbb{R}^{512}$. In addition to VGG-19, we experiment with ResNet-50 to extract features at the same spatial resolution where $a_i \in \mathbb{R}^{2048}$. Both encoders were pretrained on ImageNet [6] and kept frozen during training.

3.2 Decoder

The decoder is a LSTM that generates the caption one word at a time with teacher forcing. At each timestep t , the model predicts the next word conditioned on the previous hidden state h_{t-1} , the previously generated word y_{t-1} , and a context vector $\hat{z}_t$ derived from attention. The output is then computed using a deep output layer that combines the hidden state, context vector, and the previous word to produce a probability distribution over the vocabulary.

3.3 Attention Mechanism

At each timestep, the model computes attention weights over all spatial features:

$$e_{t,i} = f_{att}(a_i, h_{t-1}), \quad \alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_k^L \exp(e_{t,k})}.$$

Soft Attention. The context vector is computed as a weighted average $\hat{z}_t = \sum_{i=1}^L \alpha_{t,i} a_i$ multiplied by a learned gate scalar β_t . We additionally include the doubly stochastic regularization term: $\lambda_{reg} \sum_i (1 - \sum_t \alpha_{t,i})^2$, which encourages the model to attend to all regions of the image over the course of generation as the paper observed this penalty quantitatively improved BLEU score. Since this formulation is fully differentiable, we train end-to-end using backpropagation.

Hard Attention. At each timestep, a **single** location is sampled: $s_t \sim \text{Multinoulli}(\alpha_t)$, $\hat{z}_t = a_{s_t}$. Because this is not differentiable, the model is trained using the REINFORCE algorithm. The gradient estimator is:

$$\begin{aligned} \nabla L = \frac{1}{n} \sum_n \bigg[& \nabla_W \log p(y \mid \tilde{s}^n, a) \\ & + \lambda_r (R^n - b) \nabla_W \log p(\tilde{s}^n \mid a) \\ & + \lambda_e \nabla_W H[\tilde{s}^n] \bigg]. \end{aligned}$$

Here, $R^n = -\mathcal{L}_{CE}^{(n)}$ is the per-sample reward, b is an exponentially decaying moving-average baseline used to reduce variance, and H is an entropy regularization term encouraging exploration. Following the paper, with probability 0.5, the sampled attention location is replaced with its expected value α_t to further stabilize optimization.

3.4 Training

We train using teacher forcing and evaluate on the validation set after each epoch. Following the original paper, we apply dropout and perform early stopping based on BLEU-4. We experimented with both RMSProp (used in the original work) and Adam, finding comparable performance; we selected RMSProp for our final models. We reduce the learning rate by a factor of 0.5 after 2 epochs without BLEU-4 improvement. During inference, captions are generated using beam search. All hyperparameters are detailed in Table 1.

	Soft	Hard
Optimizer	RMSProp	RMSProp
Learning rate	2e-4	8e-5
Batch size	64	64
Epochs (max)	30	50
LR factor / patience	0.5 / 2	0.5 / 2
Dropout	0.5	0.5
λ_{reg}	1.0	—
Entropy coeff. λ_e	—	0.005
Beam size	7	7

Table 1: Hyperparameters used for ResNet-50 models.

3.5 Dataset

We evaluate our model on the Flickr8k dataset, which contains 8,000 images paired with five human-annotated captions each. We used the standard training, validation, and test sets provided.

4 Results & Analysis

Model	BLEU-4	METEOR
Soft-Attention VGG (Xu et al., 2015)	19.50	18.93
Soft-Attention VGG	22.61	20.72
Soft-Attention ResNet	24.20	21.66
Hard-Attention VGG (Xu et al., 2015)	21.3	20.30
Hard-Attention VGG	20.46	19.28
Hard-Attention ResNet	24.40	22.04

Table 2: BLEU-4 and METEOR metrics on Flickr8k.

Despite not having access to the hyperparameters or tokenization strategy used in the original paper, our soft attention VGG implementation exceeds the paper’s reported BLEU-4 and METEOR scores. Our hard attention VGG reimplementation, however, falls slightly short of the paper’s results. Reproducing hard attention results was considerably harder due to the high variance introduced by stochastic attention sampling and REINFORCE, making training more sensitive to hyperparameters.

Replacing ResNet-50 with VGG-19 outperforms the paper’s numbers for both attention variants. Relative to our own

VGG baselines, soft attention ResNet improves by ≈ 1.6 BLEU-4 while hard attention ResNet improves by ≈ 3.9 BLEU-4. Compared to VGG-19, the deeper residual architecture of ResNet-50 likely provides more discriminative semantic representations, enabling the attention mechanism to better align visual regions with generated words. With ResNet-50, hard attention matches soft attention in BLEU-4 and slightly exceeds it in METEOR, suggesting that the gap between the two attention variants narrows when the visual encoder is stronger. Hard attention depends on selecting a single spatial region at each timestep, making it more sensitive to the quality of encoder features; stronger ResNet representations likely make each spatial feature more semantically informative, reducing the penalty of sampling only one location.

An important contribution of the paper is that the attention mechanism provides **interpretability**. At each timestep, the attention weights $\alpha_t \in \mathbb{R}^L$ define a distribution over spatial features, which we convert into heatmaps to highlight the regions of the image the model is focusing on when generating each word. Soft attention produces continuous distributions while hard attention focuses on a spatial location (Figure 1).



A black and brown dog is playing in a yard.

A rotweiler sheppard eats a blue toy.

Figure 1: Heatmaps for soft (left) and hard (right) attention.

We can further use these visualizations to gain intuition into what the model saw for incorrect captions (Figure 2).



A man is riding a white surfboard on a ramp.

A boy in a blue shirt is riding a bike on a tire swing.

Figure 2: The model produces incorrect object predictions.

5 Reflections

Reproducing hard attention results was substantially more difficult than reproducing soft attention results as small changes in hyperparameters led to high variance and unstable training due to its stochastic nature. We also saw that encoder quality plays a significant role in performance, with stronger feature representations leading to consistent improvements across all metrics.

Future directions of this work include first, initially freezing the encoder and subsequently fine-tuning it jointly with the decoder and second, replacing the LSTM decoder with a Transformer-based architecture, which could better capture long-range dependencies, while retaining the spatial CNN encoder. Replacing the CNN encoder with a ViT which models global image relationships through self-attention may potentially improve alignment between visual regions and generated words.

The problem this paper set to solve is now better handled by modern foundation models as querying GPT, Gemini, or Claude with the same Flickr8k images produces captions that are more accurate, more natural, and more descriptive than anything our model generates. Thus, reproducing this work finally gave us a deeper appreciation for how much has changed since 2015.

References

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